

Cross Examination

Instructions

Use the simulator at https://sruckman.shinyapps.io/cross_examination/ to complete these exercises. For each scenario, read the biological question and the constraints carefully. Use the simulator to explore different mapping designs. For question 4, run the simulator twice per mapping design: once with the 1-QTL genetic model and once with the polygenic model. Save your output plots for both runs.

Answer these five questions for each scenario:

1. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.
2. What parameters would you set and why?
3. How does the practical constraint described affect your parameter choices?
4. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?
5. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.

Scenario 1: Mouse Blood Pressure

You want to map systolic blood pressure in mice. Blood pressure requires individual phenotyping by tail cuff over 8 weeks. Two resources are available. A Diversity Outbred (DO) colony derived from 8 fully intercrossed founders with 50 generations of recombination is available at \$200 per mouse, allowing 1,500 mice on your budget. A biparental F_2 cross is also feasible at \$600 per mouse, given the longer breeding time and the need for more mice before sequencing, which allows 500 F_2 mice within your budget. Your total budget is \$300,000 and applies to both approaches.

1. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.
2. What parameters would you set and why?
3. How does the practical constraint described affect your parameter choices?

4. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?

5. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.

Scenario 2: Guppy Age at Maturity

You want to map age at maturity in Trinidadian guppies, *Poecilia reticulata*, using two divergent populations from high and low predation streams. Lab crosses are feasible, but individual housing limits your total sample to 300 F_2 fish. You can rent an additional fish room, which would allow you to house a total of 2,000 F_2 fish between the two rooms.

6. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.

7. What parameters would you set and why?

8. How does the practical constraint described affect your parameter choices?

9. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?
10. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.

Scenario 3: Arabidopsis Flowering Time

You want to map flowering time in *Arabidopsis thaliana*. Two resources are available: a 14-founder NAM population with 3 generations of recombination, and an 18-founder MAGIC population with 6 generations of recombination. Flowering time is high-throughput and can be scored visually. Your growth chamber holds 1,500 plants per run. Plants take approximately 6 weeks to flower, allowing 4 sequential runs within your 6-month window. For each run, you score all 1,500 plants, select the 500 earliest-flowering as cases, and draw 500 randomly from the non-extreme plants as controls. The remaining 500 are scored but not pooled.

11. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.
12. What parameters would you set and why?
13. How does the practical constraint described affect your parameter choices?

14. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?

15. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.

Scenario 4: Yeast Fluconazole Resistance

You want to map fluconazole resistance in *Saccharomyces cerevisiae*. The 1002 Yeast Genomes project has sequenced over 1,000 natural and clinical isolates, many of which exhibit varying fluconazole minimum inhibitory concentrations (MICs). Two of the most divergent strains, one highly resistant, one fully sensitive, can be crossed in the lab. An 18-founder fully intercrossed outbred population has been maintained for 12 generations of outcrossing and supports bulk segregant analysis with effectively unlimited sample sizes. Phenotyping by spot assay on drug plates is fast and cheap for all approaches. Note: the 1002 Yeast Genomes collection contains many clonal lineages, so effective LD across the panel is longer than in humans. Set LD to 20 cM in the GWAS tab of the simulator.

16. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.

17. What parameters would you set and why?

18. How does the practical constraint described affect your parameter choices?

19. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?

20. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.

Scenario 5: Galapagos Finch Beak Depth

You want to map beak depth in Galapagos finches. These are protected wild birds; controlled crosses are not possible. 500 whole genomes have been sequenced from a small island population. Note: this population has long LD blocks. Set LD to 50 cM in the GWAS tab of the simulator.

21. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.

22. What parameters would you set and why?

23. How does the practical constraint described affect your parameter choices?

24. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?

25. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.

Scenario 6: Human Major Depressive Disorder

You want to map major depressive disorder in humans. Diagnosis relies on self-report with no objective biomarker. Biobanks have collected genotypes from 100,000 individuals. Note: humans have relatively short LD blocks. Set LD to 10 cM in the GWAS tab of the simulator.

26. Which mapping method and population would you choose, and why? If multiple resources are available, justify your choice.

27. What parameters would you set and why?

28. How does the practical constraint described affect your parameter choices?

29. Run your design with 1 QTL and with polygenic architecture. Does your method detect a signal in both cases?

30. Would a different method work here? What would you gain or lose? If no alternative is feasible, explain why.